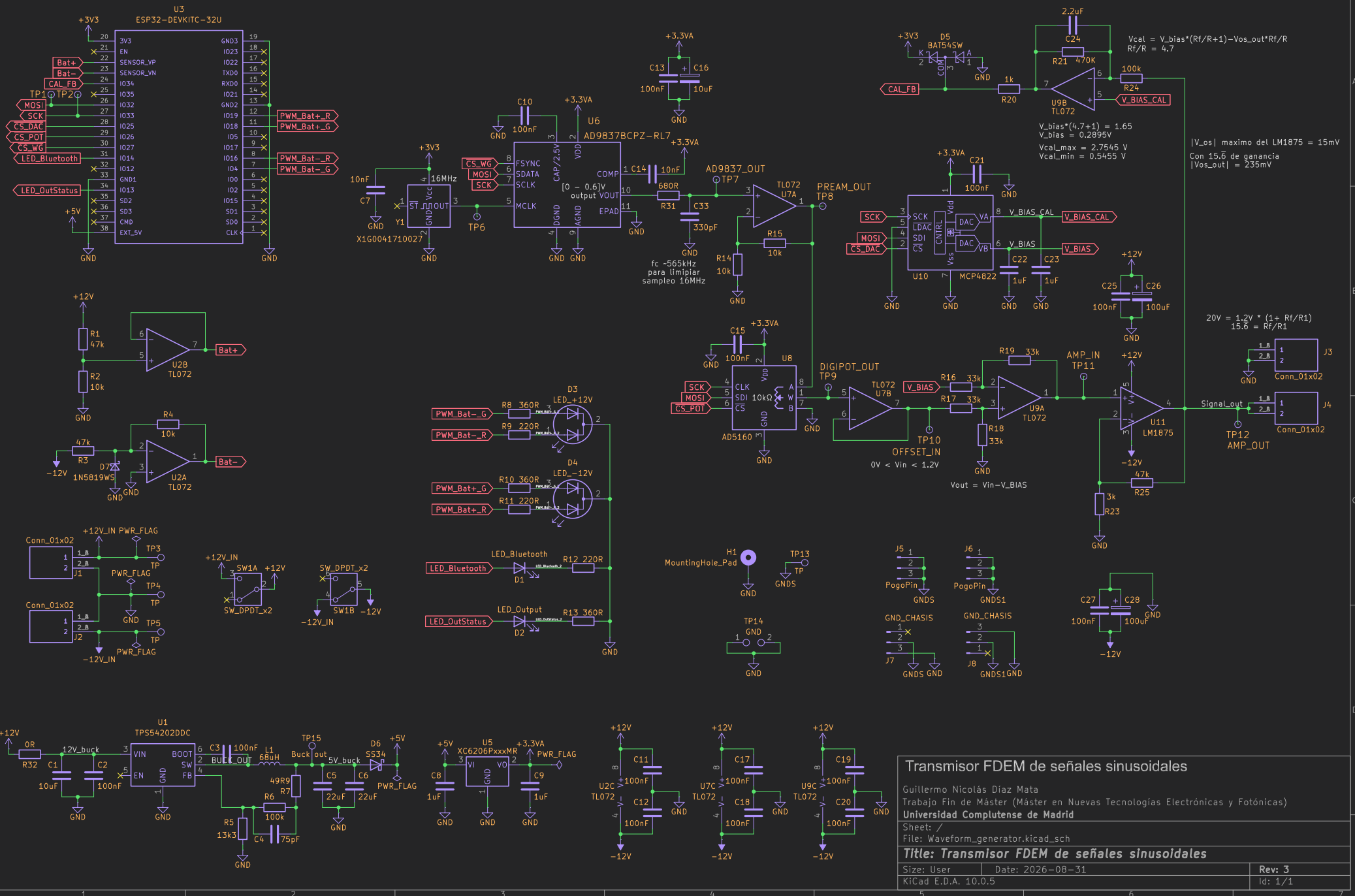

U3
ESP32-DEVKITC-32U
+3V3
Bat+
Bat-
CAL_FB
TP1
TP2
MOSI
SCK
CS_DAC
CS_POT
CS_WG
LED_Bluetooth
LED_OutStatus
+5V
GND
PWM_Bat+_R
PWM_Bat+_G
PWM_Bat-_R
PWM_Bat-_G
U6
AD9837BCPZ-RL7
C10
100nF
+3.3VA
C13
C16
10uF
10nF
C7
16MHz
Y1
X1G0041710027
TP6
FSYNC
SDATA
SCLK
MCLK
CAP/2.5V
VDD
COMP
VOUT
EPAD
DGND
AGND
C14
680R
R31
C33
330pF
AD9837_OUT
TP7
fc ~565kHz
para limpiar
sampleo 16MHz
TL072
U7A
PREAM_OUT
TP8
R15
10k
R14
10k
D5
BAT54SW
C24
2.2uF
R21
470K
R20
1k
U9B
TL072
R24
100k
V_BIAS_CAL
Vcal = V_bias*(Rf/R+1)-Vos_out*Rf/R
Rf/R = 4.7
V_bias*(4.7+1) = 1.65
V_bias = 0.2895V
Vcal_max = 2.7545 V
Vcal_min = 0.5455 V
|V_os| maximo del LM1875 = 15mV
Con 15.6 de ganancia
|Vos_out| = 235mV
C21
100nF
SCK
MOSI
CS_DAC
LDAC
SDI
CS
CNTRL
Vdd
Vss
DAC
VA
VB
V_BIAS
U10
MCP4822
C22
1uF
C23
1uF
C25
100nF
C26
100uF
+12V
20V = 1.2V * (1+ Rf/R1)
15.6 = Rf/R1
J3
J4
Conn_01x02
Signal_out
TP12
AMP_OUT
C15
100nF
U8
AD5160
10kΩ
DIGIPOT_OUT
TP9
TL072
U7B
R16
33k
R17
33k
V_BIAS
TP10
OFFSET_IN
0V < Vin < 1.2V
R19
33k
R18
33k
U9A
TL072
AMP_IN
TP11
Vout = Vin-V_BIAS
U11
LM1875
-12V
47k
R25
3k
R23
R1
47k
R2
10k
U2B
TL072
R4
10k
R3
47k
D7
1N5819WS
U2A
TL072
D3
LED_+12V
D4
LED_-12V
R8 360R
R9 220R
R10 360R
R11 220R
PWM_Bat-_G
PWM_Bat-_R
PWM_Bat+_G
PWM_Bat+_R
LED_Bluetooth
R12 220R
D1
LED_Output
R13 360R
D2
H1
MountingHole_Pad
TP13
GNDS
TP14
J5
J6
PogoPin
GNDS1
GND_CHASIS
J7
J8
GNDS GND
GNDS1GND
C27
100nF
C28
100uF
Conn_01x02
J1
J2
TP3
TP4
TP5
PWR_FLAG
+12V_IN
-12V_IN
SW_DPDT_x2
SW1A
SW1B
U1
TPS54202DDC
R32
0R
12V_buck
C1
10uF
C2
100nF
VIN
EN
BOOT
SW
FB
C3
100nF
BUCK_OUT
L1
68uH
Buck_out
5V_buck
TP15
D6
SS34
C5
22uF
C6
22uF
49R9
R7
R6
100k
R5
13k3
C4
75pF
U5
XC6206PxxxMR
VI
VO
C8
1uF
C9
1uF
U2C
U7C
U9C
TL072
C11
C12
C17
C18
C19
C20
100nF
Transmisor FDEM de señales sinusoidales
Guillermo Nicolás Díaz Mata
Trabajo Fin de Máster (Máster en Nuevas Tecnologías Electrónicas y Fotónicas)
Universidad Complutense de Madrid
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Rev: 3
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